



DIGITAL LEARNING CONTENT MANUAL

GRADE TEN

**ENGLISH
MATHS
BIOLOGY
CHEMISTRY
PHYSICS**

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Digital Content Practical Lookup Guide

1. Introduction

The Ministry of Federal Education and Professional Training is dedicated to enhancing the quality of distance learning and creating a comprehensive educational platform that serves a diverse group of learners across Pakistan. As part of this effort, the Ministry has procured digital content, including audio and video resources, that align with the Pakistan National Curriculum and Student Learning Outcomes (SLOs).

This guide is designed to help teachers navigate this digital content efficiently. Every video has been assigned a unique code, and this document organizes them in a way that clearly shows which video supports which SLO, for which grade and subject. Each table includes the video code, topic, SLO(s) covered, textbook unit, and duration.

Teachers can use this guide for lesson planning, in-class instruction, or revision sessions. It ensures that digital resources are used effectively, keeping lessons aligned with the curriculum and maximizing learning outcomes. The structure is consistent across all grades and subjects, making it easy to locate the right video at the right time.

This guide is intended as a practical companion to support teachers in delivering high-quality, curriculum-aligned instruction through digital content.

2. Available digital content from Grade KG- Grade 12

Digital Learning Content (Video)

Grade	Subject	No. of Packages	Total No. of Packages
KG	Math	21	133
	English	36	
	Urdu	25	
	General Knowledge	51	
Grade 1	Math	35	147
	English	33	
	Urdu	60	
	General Knowledge	19	
Grade 2	Math	45	166
	English	71	
	Urdu	16	
	General Knowledge	34	
Grade 3	English	92	176
	General Knowledge	29	
	Maths	39	
	Urdu	16	
Grade 4	English	92	190
	Math	36	
	Science	42	
	Urdu	20	
Grade 5	English	73	155
	Math	23	
	Science	34	
	Urdu	25	

Grade	Subject	No. of Packages	Total No. of Packages
Grade 6	Maths	31	120
	Science	27	
	English	37	
	Computer Science	25	
Grade 7	English	64	167
	Maths	51	
	Science	24	
	Computer Science	28	
Grade 8	English	60	162
	Maths	45	
	Science	25	
	Computer Science	32	

Grade	Subject	No. of Packages	Total No. of Packages
Grade 9	Biology	19	119
	Chemistry	21	
	English	20	
	Maths	42	
Grade 10	Physics	17	131
	Biology	26	
	Chemistry	12	
	English	21	
Grade 11	Maths	55	552
	Physics	17	
	Biology	110	
	Chemistry	94	
Grade 12	English	55	544
	Maths	88	
	Physics	91	
	Urdu	114	
ECD - Grade 12	Biology	103	40
	Chemistry	86	
	English	55	
	Maths	67	
ECD - Grade 12	Pak. Studies	45	40
	Physics	56	
	Urdu	132	
	Gamified Educational Content	40	

Total no. of Video Packages	2802
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Sesame, Idara Taleem o Agahi (ITA), Environment & Entrepreneurship

Sesame Workshop	
Social Emotional Learning (SEL)	17
Maths	38
Science/ General Knowledge	21
Total	76 videos 55 (5 min videos) and 21 (27 minute videos)

ITA - Storytelling	
Grade 1	6
Grade 2	6
Grade 3	7
Total	19

Theme	No. of Packages
Environments for Living Lot 1 (ECD-2)	6
Environments for Living Lot 2 (Grades 3-5)	18
Environments for Living Lot 3 (Grades 6-8)	21
Entrepreneurship (Grades 6-8)	20

3. How to read video codes

Every video in this digital content collection has a unique code. The code helps teachers quickly identify:

- the grade
- the subject
- the textbook unit
- the package and part of the video

Understanding the code allows you to find the right video quickly without searching through multiple files.

Video Code Format

Example: **G01EnU01P01a**

- G01 = Grade 1
- En = English
- U01 = Unit 1
- P01a = Package 1, Part a

Code Reference Table:

Grade	Subject	Unit	Package
G01-G12, EY	M (Mathematics)	U01	P01
G01-G12, EY	Ur (Urdu)	U01	P01
G01-G12, EY	En (English)	U01	P01
G01-G12, EY	Sc (Science)	U01	P01
G01-G12, EY	Ph (Physics)	U01	P01
G01-G12, EY	Bi (Biology)	U01	P01
G01-G12, EY	Ch (Chemistry)	U01	P01
G12	PS (Pakistan Studies)	U01	P01

Notes for Teachers:

- Use this table to decode video codes quickly.
- “EY” refers to Early Years/Kindergarten.
- Package numbers indicate video sequence within a unit.
- Always cross-check the mapping tables in this guide for SLO coverage before showing a video in class.

4. Grade 10 Digital Content Alignment with SLOs

a. English

Unit	SLO Code	SLO	Package Code	Package Title
Stopping by Woods on a Snowy Evening	1.2	Read a poem and give orally and in writing • Theme and its development. • Personal response with justification. • Paraphrase/ summary.	G10EnU01P01	Paraphrasing Poetry and Understanding Root Words
		Around the World		
	1.1	Enhance and use appropriate vocabulary and correct spelling in speech and writing; • Illustrate the use of dictionary for finding appropriate meaning and correct spellings. • Use a thesaurus to locate the synonyms closest to the meaning of the given word in the context. • Examine and interpret transitional devices that show comparison, contrast, reason, concession, condition, emphasis. • Deduce the meaning of unfamiliar words from the context using contextual clues. • Analyse and understand common roots and use that knowledge to recognize the meaning of new words. • Analyse and understand common prefixes and suffixes; use that knowledge to determine the meaning of unfamiliar words. • Understand connotations and denotations; explore the use of synonyms with varying shades of meaning used for various purposes e.g. propaganda, irony, parody and satire. • Examine and focus the vocabulary that indicates the writer's attitude. • Recognize words that vary in meaning according to their connotations. • Use appropriate connotation in their own writing. • Identify and avoid verbosity; use one word substitution, eliminate redundancy • show cultural and gender sensitivity by making appropriate adjustments in language for social, academic and daily life		

		situations. • Understand that cognates increase comprehension of spoken and written languages. Identify commonly occurring cognates in English and Urdu.		
	2.1	Famous Personalities		
	2.1	Enhance and use appropriate vocabulary and correct spelling in speech and writing; • Illustrate the use of dictionary for finding appropriate meaning and correct spellings. • Use a thesaurus to locate the synonyms closest to the meaning of the given word in the context. • Examine and interpret transitional devices that show comparison, contrast, reason, concession, condition, emphasis. • Deduce the meaning of unfamiliar words from the context using contextual clues. • Analyse and understand common roots and use that knowledge to recognize the meaning of new words. • Analyse and understand common prefixes and suffixes; use that knowledge to determine the meaning of unfamiliar words. • Understand connotations and denotations; explore the use of synonyms with varying shades of meaning used for various purposes e.g. propaganda, irony, parody and satire. • Examine and focus the vocabulary that indicates the writer's attitude. • Recognize words that vary in meaning according to their connotations. • Use appropriate connotation in their own writing. • Identify and avoid verbosity; use one word substitution, eliminate redundancy • show cultural and gender sensitivity by making appropriate adjustments in language for social, academic and daily life situations. • Understand that cognates increase comprehension of spoken and written languages. Identify commonly occurring cognates in English and Urdu.	G10EnU01P02	Prefixes, Suffixes, and Silent Letters
	2.1	Around the World		
		Enhance and use appropriate vocabulary and correct spelling in		

		speech and writing: • Illustrate the use of dictionary for finding appropriate meaning and correct spellings. • Use a thesaurus to locate the synonyms closest to the meaning of the given word in the context. • Examine and interpret transitional devices that show comparison, contrast, reason, concession, condition, emphasis. • Deduce the meaning of unfamiliar words from the context using contextual clues. • Analyse and understand common roots and use that knowledge to recognize the meaning of new words. • Analyse and understand common prefixes and suffixes; use that knowledge to determine the meaning of unfamiliar words. • Understand connotations and denotations; explore the use of synonyms with varying shades of meaning used for various purposes e.g. propaganda, irony, parody and satire. • Examine and focus the vocabulary that indicates the writer's attitude. • Recognize words that vary in meaning according to their connotations. • Use appropriate connotation in their own writing. • Identify and avoid verbosity; use one word substitution, eliminate redundancy • show cultural and gender sensitivity by making appropriate adjustments in language for social, academic and daily life situations. • Understand that cognates increase comprehension of spoken and written languages. Identify commonly occurring cognates in English and Urdu.		
	2.1	Recognize silent letters in words and pronounce them with developing accuracy.		
A Mother's Letter	2.1	Apply critical thinking to interact with text and use intensive reading strategies (while-reading) to • scan to answer short questions. • make simple inferences using context of the text and prior knowledge. • distinguish between what is clearly stated and what is implied . • deduce meaning of	G10EnU02P01	Analysing Text and Using Idioms

		difficult words from context. • use context to infer missing words. • read silently with comprehension and extract main idea and supporting detail. • scan to locate an opinion. • distinguish fact from opinion in letters to editors, texts supporting an opinion, etc. • locate examples to support an opinion e.g. a personal anecdote, quotations, examples and analogy, etc. • recognize arguments and counter arguments. • explore viewpoints/ ideas and issues. • follow instructions in maps or user instruction manuals and forms related to extended social and academic environment. • comprehend/interpret text by applying critical thinking . • generate questions to understand text.		
		Around the World		
	2.1	Translate passages from English to Urdu: • Use the knowledge of literal and figurative meaning, grammatical gender and syntax to translate passages from English to Urdu. • Understand that most phrases and idioms do not translate literally from one language to another.		
	1.4	Apply critical thinking to interact with text and use intensive reading strategies (while-reading) to • scan to answer short questions. • make simple inferences using context of the text and prior knowledge. • distinguish between what is clearly stated and what is implied . • deduce meaning of difficult words from context. • use context to infer missing words. • read silently with comprehension and extract main idea and supporting detail. • scan to locate an opinion. • distinguish fact from opinion in letters to editors, texts supporting an opinion, etc. • locate examples to support an opinion e.g. a personal anecdote, quotations, examples and analogy, etc. • recognize arguments and counter arguments. • explore viewpoints/ ideas and issues. • follow instructions in maps or user instruction manuals	G10EnU02P02	Analysing Text and Making Adjectives

		and forms related to extended social and academic environment. • comprehend/interpret text by applying critical thinking . • generate questions to understand text.		
		Around the World		
	2.1	Form adjectives from nouns and verbs.		
	2.1	Famous Personalities		
Terror in the Night	1.2	Read text to make predictions about story line / content, characters, using contextual clues and prior knowledge.	G10EnU03P01	Stories, Proverbs, and Presentations
		Around the World		
	1.1	Translate passages from English to Urdu: • Use the knowledge of literal and figurative meaning, grammatical gender and syntax to translate passages from English to Urdu. • Understand that most phrases and idioms do not translate literally from one language to another.		
		Famous Personalities		
	2.1	Create and deliver simple group/ class presentations on various themes, problems and issues	G10EnU03P02	Adjectives, Outlines, and Interviews
	2.1	Use adjective phrases and clauses.		
	2.1	Use adjective phrases and clauses.		
		Around the World		
	1.1	Select and use a variety of pre-writing strategies such as brainstorming, mind mapping, outlining etc.		
		Famous Personalities		
	2.1	Identify and analyse appropriate expressions and interviewing techniques to conduct and give short interviews of a friend or relative on specific topics		
The Raven	1.1	Recognize genres of literature, fiction, nonfiction, poetry, legend, myth.	G10EnU04P01	Fiction and First Conditionals
		Around the World		
	2.1	Use conditional sentences.		
	1.4	Use library skills to • alphabetize book titles, words and names. • locate fiction and non fiction books / books by subject. • understand card catalogue. • locate and use card catalogue. • identify three kinds of catalogue cards i.e. author card, title card, subject card. • use case and shelf labels in the	G10EnU04P02	Library Skills, Infinitives, and Colons

		library. • use the reference section in the library. • use computer catalogue.		
		Around the World		
	2.1	Illustrate use of infinitives and infinitive phrases.		
		Famous Personalities		
	2.2	Recognize and use semicolon between • parts of a compound sentence when no conjunction is used. • the clauses of a compound sentence before conjunctive adverbs.		
An Honest Young Man (A Folk Tale)	2.1	Recognize genres of literature, fiction, nonfiction, poetry, legend, myth.	G10EnU05P01	Folktales, Conditionals, and Intonation
		Around the World		
	2.1	Use conditional sentences.		
		Hamari Duniya		
	2.1	Recognize and use varying intonation patterns as aids in spoken and written discourse	G10EnU05P02	Inferences and Perfect Continuous Tenses
	2.1	Read a text to • make connections between their own lives and the characters, events, motives, and causes of conflict in texts. • make inferences and draw conclusion about character using supportive evidence from the text . • compare characters in a literary selection to near similar one's in real life.		
		Around the World		
	2.1	Identify, change the form of, and use Present and Past Perfect Continuous tenses.		
	2.1	Identify, change the form of, and use Present and Past Perfect Continuous tenses.		
The Fantastic Shoemaker	2.1	Identify the speaker or narrator in a selection.	G10EnU06P01	Speaking and Pronunciation
		Around the World		
	1.1	Use the pronunciation key (in a dictionary) to pronounce words with developing accuracy.		
	1.3	Recognize, pronounce and represent primary and secondary stress in words. Refer to a dictionary, if required.		
	2.1	Use adverbial phrases and clauses.	G10EnU06P02	Adverbs and Opinion Editorials
		Famous Personalities		
	2.1	Use adverbial phrases and clauses.		
		Around the World		

	1.3	Analyse letters to editor to recognize effective ways of explaining and presenting an idea, a point, or an opinion.		
A Daring Village Rescue	2.1	Recognize the author's purpose and point of view.	G10EnU07P01	Conditionals and Points of View
		Famous Personalities		
	2.3	Use conditional sentences.		
		Around the World		
	Present and explain one's point of view clearly	Understand the Conventions of Explaining One's Point of View		
	2.2	Use critical thinking to respond orally and in writing to the text (post-reading) to • give a personal opinion and justify stance related to viewpoints/ ideas and issues in the text read. • relate what is read to his or her own feelings and experiences . • explore causes and consequences of a problem or an issue and propose various solutions. • evaluate material read.	G10EnU07P02	Present Perfect and Negotiations
		Famous Personalities		
	2.1	Illustrate use of tenses learnt earlier.		
		Around the World		
	1.1	Use polite forms to negotiate and reach consensus		
Oliver Twist	2.2	Read a text to • make connections between their own lives and the characters, events, motives, and causes of conflict in texts. • make inferences and draw conclusion about character using supportive evidence from the text . • compare characters in a literary selection to near similar one's in real life.	G10EnU08P01	Inferences, Conclusions, and Summaries
		Hamari Duniya		
	1.2	Use summary skills to • extract salient points and develop a mind map to summarize a text. • follow a process or procedure to summarize the information to transfer the written text to a diagram flow chart or cloze paragraph.		
	1.1	Write an essay on a general subject: • Write an introductory paragraph with a clear central thought. • Provide key	G10EnU08P02	Expository Essay

		ideas which prove, explain or support the central thought. • Use a separate paragraph for each key idea. • Incorporate evidence (facts, quotations, etc), examples (analogies, anecdotes, etc.), or different points of view (elaborating an idea/opinion) to support each key idea. • Use appropriate transitional devices to connect ideas within and between paragraphs. • Add a closing or summary paragraph with a synthesis of central idea, syntheses of each supporting idea, a general concluding statement. • Use correct conventions of grammar and punctuation. • Use appropriate vocabulary.		
		Around the World		
King For a Day	1.2	Use summary skills to • extract salient points and develop a mind map to summarize a text. • follow a process or procedure to summarize the information to transfer the written text to a diagram flow chart or cloze paragraph.	G10EnU09P01	Summaries and Essays
		Around the World		
	1.1	Write an essay on a general subject: • Write an introductory paragraph with a clear central thought. • Provide key ideas which prove, explain or support the central thought. • Use a separate paragraph for each key idea. • Incorporate evidence (facts, quotations, etc), examples (analogies, anecdotes, etc.), or different points of view (elaborating an idea/opinion) to support each key idea. • Use appropriate transitional devices to connect ideas within and between paragraphs. • Add a closing or summary paragraph with a synthesis of central idea, syntheses of each supporting idea, a general concluding statement. • Use correct conventions of grammar and punctuation. • Use appropriate vocabulary.		
	1.1	Write an essay on a general subject: • Write an introductory paragraph with a clear central thought. • Provide key ideas which prove, explain or support	G10EnU09P02	Essays, Hyphens, and Dashes

		the central thought. • Use a separate paragraph for each key idea. • Incorporate evidence (facts, quotations, etc), examples (analogies, anecdotes, etc.), or different points of view (elaborating an idea/opinion) to support each key idea. • Use appropriate transitional devices to connect ideas within and between paragraphs. • Add a closing or summary paragraph with a synthesis of central idea, syntheses of each supporting idea, a general concluding statement. • Use correct conventions of grammar and punctuation. • Use appropriate vocabulary.		
		Hamari Duniya		
	2.2	Recognize and use hyphen to indicate the division of a word at the end of a line. Recognize and use dash as a separator to indicate that a sentence has been broken off, or to indicate a new direction of thought.		
Greatest Scientific Discoveries	1.4	Use textual aids such as table of contents, footnote, glossary, preface, sub headings etc. to • comprehend texts. • identify and select relevant information in a book.	G10EnU10P01	Paragraphs, Connotation, and Denotation
		Around the World		
	1.1	Analyse an essay to identify the general subject, main idea (a statement about the general subject), key ideas, supporting details and transitional devices.		
		Famous Personalities		
	1.1	Enhance and use appropriate vocabulary and correct spelling in speech and writing: • Illustrate the use of dictionary for finding appropriate meaning and correct spellings. • Use a thesaurus to locate the synonyms closest to the meaning of the given word in the context. • Examine and interpret transitional devices that show comparison, contrast, reason, concession, condition, emphasis. • Deduce the meaning of unfamiliar words from the context using contextual clues. • Analyse and understand common roots and use		

		that knowledge to recognize the meaning of new words. • Analyse and understand common prefixes and suffixes; use that knowledge to determine the meaning of unfamiliar words. • Understand connotations and denotations; explore the use of synonyms with varying shades of meaning used for various purposes e.g. propaganda, irony, parody and satire. • Examine and focus the vocabulary that indicates the writer's attitude. • Recognize words that vary in meaning according to their connotations. • Use appropriate connotation in their own writing. • Identify and avoid verbosity; use one word substitution, eliminate redundancy • show cultural and gender sensitivity by making appropriate adjustments in language for social, academic and daily life situations. • Understand that cognates increase comprehension of spoken and written languages. Identify commonly occurring cognates in English and Urdu.		
	E-06-C4-05	Use past continuous in writing and past perfect for a past action occurring before another past action (e.g., I had done my homework before my father reached home)	G10EnU10P02	Past Perfect, Punctuation, and Letters
	2.1	Illustrate use of punctuation marks learnt earlier.		
		Hamari Duniya		
	1.3	Analyse and compare various informal and formal emails to note differences of conventions, vocabulary, style and tone.		
	2.2	Recognize and use parenthesis (Round Brackets) to • enclose numbers or letters in enumerations in the text. • express an amount in numbers previously expressed in words. • mark off explanatory or supplementary material. Recognize and use omission marks or ellipses to signify the omission or deletion of letters or words in sentences.	G10EnU10P03	Sentences, Ellipses, and Brackets

	2.3	Identify and differentiate between simple, compound and complex sentences.		
		Famous Personalities		
	2.3	Identify and differentiate between simple, compound and complex sentences.		
	2.3	Identify and differentiate between simple, compound and complex sentences.		

b. Maths

Unit Name	Learning Outcomes /Skills	Skill Name	Package Code	Package Title
QUADRATIC EQUATIONS	Define quadratic equation.	Define quadratic equation	G10MU01P01	Solving Quadratic Equations Using Factorisation
	Solve a quadratic equation in one variable by - factorization, - completing square.	Solve a quadratic equation in one variable by factorization		
		Solve a quadratic equation in one variable by completing square	G10MU01P02	Solving Quadratic Equations
	i) Use method of completing square to derive quadratic formula.	Derive quadratic formula by using method of completing square		
	ii) Use quadratic formula to solve quadratic equations.	Solve a quadratic equation by using quadratic formula		
	Solve the equation of the type $ax^2 + bx + c = 0$ by reducing it to the quadratic form	Solve the equation of the type $ax^2 + bx + c = 0$ by reducing it to the quadratic form	G10MU01P03	Solve Equations by Reducing Them to Quadratic Form
	Solve the equation of the type $a p(x) + b / q(x) = c$	Solve the equation of the type $a p(x) + b / q(x) = c$		
	Solve the reciprocal equations of the type $a (x^2 + 1/x^2) + b (x + 1/x) + c = 0$	Solve the reciprocal equations of the type $a (x^2 + 1/x^2) + b (x + 1/x) + c = 0$		
	Solve exponential equations involving variables in exponents	Solve exponential equations involving variables in exponents	G10MU01P04	Solving Equations Reducible to Quadratic Form
	Solve equations of the type $(x + a) (x + b) (x + c) (x + d) = k$, where $a + b = c + d$	Solve equations of the type $(x + a) (x + b) (x + c) (x + d) = k$, where $a + b = c + d$		
	Solve the radical equation of the type: $\sqrt{ax + b} = cx + d$	Solve the radical equation of the type: $\sqrt{ax + b} = cx + d$	G10MU01P05	Solve Radical Equations
	Solve the radical equation of the type: $\sqrt{x + a} + \sqrt{x + b} = \sqrt{x + c}$	Solve the radical equation of the type: $\sqrt{x + a} + \sqrt{x + b} = \sqrt{x + c}$		

	Solve the radical equation of the type: $\sqrt{x^2 + px + m} + \sqrt{x^2 + px + n} = q$	Solve the radical equation of the type: $\sqrt{x^2 + px + m} + \sqrt{x^2 + px + n} = q$		
THEORY OF QUADRATIC EQUATIONS	i) Define discriminant ($b^2 - 4ac$) of the quadratic expression $ax^2 + bx + c$ ii) Find discriminant of a given quadratic equation.	Define discriminant ($b^2 - 4ac$) of the quadratic expression $ax^2 + bx + c$, and find discriminant of a given quadratic equation	G10MU02P01	Nature of Roots of a Quadratic Equation
	iii) Discuss the nature of roots of a quadratic equation through discriminant.	Discuss the nature of roots of a quadratic equation through discriminant		
	iv) Determine the nature of roots of a given quadratic equation and verify the result by solving the equation.	Determine the nature of roots of a given quadratic equation and verify the results by solving the equation		
	v) Determine the value of an unknown involved in a given quadratic equation when the nature of its roots is given.	Determine the value of an unknown involved in a given quadratic equation when the nature of its roots is given		
	i) Find cube roots of unity. ii) Recognize complex cube roots of unity as ω and ω^2 .	Find cube roots of unity, and recognise complex cube roots of unity as ω and ω^2	G10MU02P02	Cube Roots of Unity and Their Properties
	iii) Prove the properties of cube roots of unity.	Prove the properties of cube roots of unity		
	iv) Use properties of cube roots of unity to solve appropriate problems.	Use properties of cube roots of unity to solve appropriate problems		
	i) Find the relation between the roots and the coefficients of a quadratic equation. ii) Find the sum and product of roots of a given quadratic equation without solving it.	Find the relation between the roots and the coefficients of a quadratic equation, and find the sum and products of the roots of a quadratic equation without solving it	G10MU02P03	Finding Unknowns in Quadratic Equations

	iii) Find the value(s) of unknown(s) involved in a given quadratic equation when • sum of roots is equal to a multiple of the product of roots,	Find the values of unknowns involved in a given quadratic equation when sum of roots is equal to a multiple of the products of roots		
	• sum of the squares of roots is equal to a given number,	Find the values of unknowns involved in a given quadratic equation when sum of the squares of roots is equal to a given number		
	• roots differ by a given number,	Find the values of unknowns involved in a given quadratic equation when roots differ by a given number		
	• roots satisfy a given relation (e.g. the relation $2\alpha + 5\beta = 7$ where α and β are the roots of given equation),	Find the values of unknowns involved in a given quadratic equation when roots satisfy a given relation (Eg. $2a + 5b = 7$, where a and b are roots)	G10MU02P04	Find Unknowns in Quadratic Equations and Symmetric Functions
	• both sum and product of roots are equal to a given number.	Find the values of unknowns involved in a given quadratic equation when both sum and products of roots are equal to a given number		
	i) Define symmetric functions of roots of a quadratic equation.	Define symmetric functions of roots of a quadratic equation		
	ii) Evaluate a symmetric function of the roots of a quadratic equation in terms of its coefficients.	Evaluate a symmetric function of the roots of a quadratic equation in terms of its coefficients		
	Establish the formula, $x^2 - (\text{sum of roots})x + (\text{products of roots}) = 0$ to find a quadratic equation from the given roots	Establish the formula, $x^2 - (\text{sum of roots})x + (\text{products of roots}) = 0$ to find a quadratic equation from the given roots	G10MU02P05	Formation of Quadratic Equations Part 1

	Form the quadratic equation whose roots are of the type: $2a + 1$; $2b + 1$, where a and b are roots of the quadratic equation	Form the quadratic equation whose roots are of the type: $2a + 1$; $2b + 1$, where a and b are roots of the quadratic equation		
	Form the quadratic equation whose roots are of the type: a^2 ; b^2 , where a and b are roots of the quadratic equation	Form the quadratic equation whose roots are of the type: a^2 ; b^2 , where a and b are roots of the quadratic equation		
	Form the quadratic equation whose roots are of the type: $1/a$; $1/b$, where a and b are roots of the quadratic equation	Form the quadratic equation whose roots are of the type: $1/a$; $1/b$, where a and b are roots of the quadratic equation		
	Form the quadratic equation whose roots are of the type: a/b ; b/a , where a and b are roots of the quadratic equation	Form the quadratic equation whose roots are of the type: a/b ; b/a , where a and b are roots of the quadratic equation	G10MU02P06	Formation of Quadratic Equations Part 2
	Form the quadratic equation whose roots are of the type: $a + b$; $1/a + 1/b$, where a and b are roots of the quadratic equation	Form the quadratic equation whose roots are of the type: $a + b$; $1/a + 1/b$, where a and b are roots of the quadratic equation		
	i) Describe the method of synthetic division. ii) Use synthetic division to • find quotient and remainder when a given polynomial is divided by a linear polynomial,	Describe the method of synthetic division, and use it to find the quotient and remainder when a given polynomial is divided by a linear polynomial	G10MU02P07	Synthetic Division
	• find the value(s) of unknown(s) if the zeros of a polynomial are given,	Use synthetic division to find the value(s) of unknowns if the zeros of a polynomial are given		

	find the value(s) of unknown(s) if the factors of a polynomial are given,	Use synthetic division to find the value(s) of unknowns if the factors of a polynomial are given.		
	solve a cubic equation if one root of the equation is given,	Use synthetic division to solve a cubic equation if one root of the equation is given	G10MU02P08	Solving Equations by Using Synthetic Division
	solve a biquadratic (quartic) equation if two of the real roots of the equation are given.	Use synthetic division to solve a biquadratic equation if two of the real roots of the equation are given		
	Solve a system of two equations in two variables when • one equation is linear and the other is quadratic,	Solve a system of two equations in two variables when one equation is linear and the other one is quadratic	G10MU02P09	Simultaneous Equations
	Solve the real life problems leading to quadratic equations.	Solve real life problems leading to quadratic equations		
	both the equations are quadratic.	Solve a system of two equations in two variables when both the equations are quadratic	G10MU02P10	Solving Simultaneous Equations
VARIATIONS	i) Define ratio, proportions and variations (direct and inverse).	Define ratio, proportions and variations (direct and inverse)	G10MU03P01	Ratios and Proportions
	ii) Find 3rd, 4th mean and continued proportion.	Find the 3rd, 4th, and mean proportional	G10MU03P02	Theorems on Proportions and Joint Variation
	Apply theorems of invertendo, alternendo, componendo, dividendo and componendo & dividendo to find proportions.	Apply theorems of invertendo, alternendo, componendo, dividendo and componendo & dividendo to find proportions		
	i) Define joint variation. ii) Solve problems related to joint variation.	Define joint variation and Solve		

		problems related to joint variation		
	i) Use K-Method to prove conditional equalities involving proportions.	Use k-method to prove conditional equations involving proportions	G10MU03P03	Proportions and K-Method
	ii) Solve real life problems based on variations.	Solve real life problems based on variations		
PARTIAL FRACTIONS	Define proper, improper and rational fraction.	Define proper, improper and rational fractions	G10MU04P01	Partial Fractions Part 1
	Resolve an algebraic fraction into partial fractions when its denominator consists of • non-repeated linear factors,	Resolve an algebraic fraction into partial fractions when its denominator consists of non-repeated linear factors		
	• repeated linear factors,	Resolve an algebraic fraction into partial fractions when its denominator consists of repeated linear factors	G10MU04P02	Partial Fractions Part 2
	• repeated quadratic factors.	Resolve an algebraic fraction into partial fractions when its denominator consists of repeated quadratic factors		
	• non-repeated quadratic factors,	Resolve an algebraic fraction into partial fractions when its denominator consists of non-repeated quadratic factors	G10MU04P03	Partial Fractions Part 3
SETS AND FUNCTIONS	i) Recall the sets denoted by N, Z, W, E, O, P and Q.	Recall important sets denoted by N, W, Z, E, O, P, Q and R	G10MU05P01	Operations on Sets
	ii) Recognize operation on sets ($\cup$, $\cap$, $\setminus$, K). iii) Perform operation on sets • union, • intersection, • difference, • complement.	Recognise and perform operation on sets: (Union, Intersection, Difference and Complement)		

	iv) Give formal proofs of the following fundamental properties of union and intersection of two or three sets. • Commutative property of union, • Associative property of union,	Give formal proof and verification of the commutative and associative properties of union	G10MU05P02	Properties of Union and Intersection Part 1
	• Commutative property of intersection, • Associative property of intersection,	Give formal proof and verification of the commutative and associative properties of intersection	G10MU05P03	Properties of Union and Intersection Part 2
	• Distributive property of union over intersection,	Give formal proof and verification of the distributive property of union over intersection		
	• Distributive property of intersection over union,	Give formal proof and verification of the distributive property of intersection over union	G10MU05P04	Properties of Union and Intersection Part 3
	• De Morgan's laws.	Give formal proof and verification of De-Morgan's laws	G10MU05P05	De Morgan's Laws
	vi) Use Venn diagram to represent • union and intersection of sets, • complement of a set.	Use Venn diagram to represent union, intersection and complement of sets	G10MU05P06	Venn Diagrams Part 1
	vii) Use Venn diagram to verify • commutative law for union and intersection of sets,	Use Venn diagram to verify commutative and associative laws of union		
	• associative laws,	Use Venn diagram to verify commutative and associative laws of intersection	G10MU05P07	Venn Diagrams Part 2
	• distributive laws.	Use Venn diagram to verify distributive laws		
	• De Morgan's laws,	Use Venn diagram to verify De Morgan's laws		

	viii) Recognize ordered pairs and Cartesian product.	Recognise ordered pairs and cartesian product	G10MU05P08	Ordered Pairs, Binary Relations, and Functions
	Define binary relation and identify its domain and range.	Define binary relation and identify its domain and range		
	i) Define function and identify its domain, co-domain and range.	Define function and identify its domain, co-domain and range		
	ii) Demonstrate the following: • into function,	Demonstrate: into function	G10MU05P09	Functions
	• one-one function, • into and one-one function (injective function),	Demonstrate: one-one function (injective function)		
	• onto function (surjective function),	Demonstrate: onto functions (surjective function)		
	• one-one and onto function (bijective function).	Demonstrate: one-one and onto function (bijective function)	G10MU05P10	Different Types of Functions
	iii) Examine whether a given relation is a function or not.	Examine whether a given relation is a function or not		
	iv) Differentiate between one-one correspondence and one-one function.	Difference between one-one correspondence and one-one function		
	v) Include sufficient exercises to clarify/differentiate between the above concepts.	Include sufficient exercises to classify/differentiate between the above concepts		
BASIC STATISTICS	i) Construct grouped frequency table.	Construct grouped frequency table	G10MU06P01	Frequency Distribution
	ii) Construct histograms with equal and unequal class intervals.	Construct histograms with equal and unequal class intervals		
	iii) Construct a frequency polygon.	Construct a frequency polygon		
	i) Construct a cumulative frequency table.	Construct a cumulative frequency table	G10MU06P02	Cumulative Frequency Distribution
	ii) Draw a cumulative frequency polygon.	Draw a cumulative frequency polygon		

	i) Calculate (for ungrouped and grouped data): • arithmetic mean by definition and using deviations from assumed mean,	Calculate arithmetic mean by definition and using deviation from assumed mean (grouped and ungrouped)	G10MU06P03	Arithmetic Mean
	• median, mode, geometric mean, harmonic mean.	Calculate median, mode, geometric mean and harmonic mean	G10MU06P04	Measures of Central Tendency
	ii) Recognize properties of arithmetic mean.	Recognise properties of arithmetic mean		
	iii) Calculate weighted mean and moving averages.	Calculation of weighted mean and moving averages		
	iv) Estimate median, quartiles and mode graphically.	Estimate median, quartiles and mode graphically	G10MU06P05	Measures of Central Tendency and Dispersion
	Measure range, variance and standard deviation.	Measure range, variance and standard deviation		
INTRODUCTION TO TRIGONOMETRY	i) Measure an angle in sexagesimal system (degree, minute and second).	Measure an angle in degree, minute and second	G10MU07P01	Measurement of an Angle
	ii) Convert an angle given in Do M'S'' form into a decimal form and vice versa.	Convert an angle given in degrees, minutes and seconds into decimal form and vice versa		
	iii) Define a radian (measure of an angle in circular system) and prove the relationship between radians and degrees.	Define a radian (measure of an angle in circular system) and prove the relationship between radians and degrees		
	i) Establish the rule $l = r\theta$, where r is the radius of the circle, l the length of circular arc and θ the central angle measured in radians.	To establish and apply the formula for the length of a circular arc $l = r\theta$	G10MU07P02	Sector of a Circle and Coterminal Angles
	ii) Prove that the area of a sector of a circle is $\frac{1}{2} r^2 \theta$	To establish and apply the formula for the area of a circular sector $\frac{1}{2} r^2 \theta$		

	i) Define and identify: • general angle (coterminal angles),	Define and identify general angle (coterminal angles)		
	• angle in standard position.	Define and identify angle in standard position		
	ii) Recognize quadrants and quadrantal angles.	Recognise quadrants and quadrantal angles	G10MU07P03	Quadrants and Quadrantal Angles
	iii) Define trigonometric ratios and their reciprocals with the help of a unit circle.	Define trigonometric ratios and their reciprocals with the help of a unit circle		
	iv) Recall the values of trigonometric ratios for 45° , 30° , 60° .	Recall the values of trigonometric ratios for 45° , 30° and 60°	G10MU07P04	Trigonometric Ratios
	v) Recognize signs of trigonometric ratios in different quadrants.	Recognise signs of trigonometric ratios in different quadrants		
	vi) Find the values of remaining trigonometric ratios if one trigonometric ratio is given.	Find values of remaining trigonometric ratios if one trigonometric ratio is given		
	vii) Calculate the values of trigonometric ratios for 0° , 90° , 180° , 270° , 360° .	Calculate the values of trigonometric ratios for 0° , 90° , 180° , 270° and 360°	G10MU07P05	Trigonometric Ratios and Identities
	Prove the trigonometric identities and apply them to show different trigonometric relations.	Prove the trigonometric identities and apply them to show different trigonometric relations		
	i) Find angle of elevation and depression. ii) Solve real life problems involving angle of elevation and depression.	Find angles of elevation and depression		
PROJECTION OF A SIDE OF A TRIANGLE	Prove the following theorems along with corollaries and apply them to solve appropriate problems. i) In an obtuse-angled triangle, the square on the side opposite to the obtuse angle is equal to the	Prove and apply triangle theorem 1: $(BC)^2 = (AC)^2 + (AB)^2 + 2(AB)(AD)$	G10MU08P01	Projection of a Side of a Triangle

	sum of the squares on the sides containing the obtuse angle together with twice the rectangle contained by one of the sides, and the projection on it of the other.			
	ii) In any triangle, the square on the side opposite to an acute angle is equal to the sum of the squares on the sides containing that acute angle diminished by twice the rectangle contained by one of those sides and the projection on it of the other.	Prove and apply triangle theorem 2: $(BC)^2 = (AC)^2 + (AB)^2 - 2(AB)(AD)$		
	iii) In any triangle, the sum of the squares on any two sides is equal to twice the square on half the third side together with twice the square on the median which bisects the third side (Apollonius' Theorem).	Prove and apply Apollonius' theorem: $(AB)^2 + (AC)^2 = 2(BD)^2 + 2(AD)^2$		
CHORDS OF A CIRCLE	Prove the following theorems along with corollaries and apply them to solve appropriate problems. i) One and only one circle can pass through three non collinear points.	Prove and apply: One and only one circle can pass through three non-collinear points	G10MU09P01	Chords of a Circle Part 1
	ii) A straight line, drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord. iii) Perpendicular from the centre of a circle on a chord bisects it.	Prove and apply: A straight line, drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord (and vice versa)		
	iv) If two chords of a circle are congruent then they will be equidistant from the centre. v) Two chords of a circle which are equidistant from the centre are congruent.	Prove and apply: If two chords of a circle are congruent then they will be equidistant from the centre (and vice versa)	G10MU09P02	Chords of a Circle Part 2
TANGENT TO A CIRCLE	Prove the following theorems along with	Prove and apply: If a line is drawn	G10MU10P01	Theorem Related to

	corollaries and apply them to solve appropriate problems. i) If a line is drawn perpendicular to a radial segment of a circle at its outer end point, it is tangent to the circle at that point. ii) The tangent to a circle and the radial segment joining the point of contact and the centre are perpendicular to each other.	perpendicular to a radial segment of a circle at its outer end point, it is tangent to the circle at that point (and vice versa)		Tangent to a Circle
	iii) The two tangents drawn to a circle from a point outside it, are equal in length.	Prove and apply: The two tangents drawn to a circle from a point outside it, are equal in length		
	iv) If two circles touch externally or internally, the distance between their centres is respectively equal to the sum or difference of their radii.	Prove and apply: If two circles touch externally or internally, the distance between their centers is respectively equal to the sum or difference of their radii		
CHORDS AND ARCS	Prove the following theorems along with corollaries and apply them to solve appropriate problems. i) If two arcs of a circle (or of congruent circles) are congruent then the corresponding chords are equal. ii) If two chords of a circle (or of congruent circles) are equal, then their corresponding arcs (minor, major or semi-circular) are congruent.	If two arcs of a circle (or of congruent circles) are congruent, then the corresponding chords are equal (and vice versa)	G10MU11P01	Chords and Arcs
	iii) Equal chords of a circle (or of congruent circles) subtend equal angles at the centre (at the corresponding centres). iv) If the angles subtended by two chords of a circle (or congruent circles) at the	Equal chords of a circle (or of congruent circles) subtend equal angles at the corresponding centres (and vice versa)		

	centre (corresponding centres) are equal, the chords are equal.			
ANGLE IN A SEGMENT OF A CIRCLE	Prove the following theorems along with corollaries and apply them to solve appropriate problems. i) The measure of a central angle of a minor arc of a circle, is double that of the angle subtended by the corresponding major arc.	The measure of a central angle of a minor arc of a circle, is double that of the angle subtended by the corresponding major arc	G10MU12P01	Angles in a Segment of a Circle Part 1
	ii) Any two angles in the same segment of a circle are equal.	Any two angles in the same segment of a circle are equal		
	iii) The angle $\angle$ in a semi-circle is a right angle,	The angle in a semicircle is a right angle		
	$\angle$ in a segment greater than a semi circle is less than a right angle, $\angle$ in a segment less than a semi-circle is greater than a right angle,	The angle in a segment greater than a semi circle is less than a right angle, and vice versa		
	iv) The opposite angles of any quadrilateral inscribed in a circle are supplementary.	The opposite angles of any quadrilateral inscribed in a circle are supplementary	G10MU12P02	Angles in a Segment of a Circle Part 2
PRACTICAL GEOMETRY – CIRCLES	i) Locate the centre of a given circle.	Locate the centre of a given circle	G10MU13P01	Construction of Circles
	ii) Draw a circle passing through three given noncollinear points.	Draw a circle passing through three given non-collinear points		
	iii) Complete the circle: $\odot$ by finding the centre,	Complete the circle when a part of its circumference is given: by finding its centre		
	$\odot$ without finding the centre, when a part of its circumference is given.	Complete the circle when a part of its circumference is given: without finding its centre		
	i) Circumscribe a circle about a given triangle.	Circumscribe a circle about a given triangle	G10MU13P02	Circles Attached to Triangles Part 1
	ii) Inscribe a circle in a given triangle.	Inscribe a circle in a given triangle		
	iii) Escribe a circle to a given triangle.	Escribe a circle in a given triangle		

	iv) Circumscribe an equilateral triangle about a given circle	Circumscribe an equilateral triangle about a given circle	G10MU13P03	Circles Attached to Triangles Part 2
	v) Inscribe an equilateral triangle in a given circle.	Inscribe an equilateral triangle in a given circle		
	vi) Circumscribe a square about a given circle.	Circumscribe a square about a given circle	G10MU13P04	Circles Attached to Squares and Hexagons
	vii) Inscribe a square in a given circle.	Inscribe a square in a given circle		
	viii) Circumscribe a regular hexagon about a given circle.	Circumscribe a regular hexagon about a given circle		
	ix) Inscribe a regular hexagon in a given circle.	Inscribe a regular hexagon in a given circle		
	i) Draw a tangent to a given arc, without using the centre, through a given point P when P is • the middle point of the arc, • at the end of the arc, • outside the arc.	Draw a tangent to an arc through a given point without using the centre	G10MU13P05	Tangent to a Circle Part 1
	ii) Draw a tangent to a given circle from a point P when P lies • on the circumference, • outside the circle.	Draw a tangent to a circle from a point on or outside the circle		
	iii) Draw two tangents to a circle meeting each other at a given angle.	Draw two tangents to a circle meeting each other at a given angle	G10MU13P06	Tangent to a Circle Part 2
	iv) Draw • direct common tangent or external tangent, • transverse common tangent or internal tangent to two equal circles.	Draw direct or transverse common tangents to two equal circles		
	v) Draw • direct common tangent or external tangent, • transverse common tangent or internal tangent to two unequal circles.	Draw direct or transverse common tangents to two unequal circles	G10MU13P07	Tangent to a Circle Part 3
	vi) Draw a tangent to • two unequal touching circles, • two unequal intersecting circles.	Draw a tangent to two unequal touching or intersecting circles	G10MU13P08	Tangent to a Circle Part 4

	vii) Draw a circle which touches • both the arms of a given angle,	Draw a circle touching both the arms of a given angle		
	• two converging lines and passes through a given point between them, • three converging lines.	Draw a circle touching two converging lines and a point between them; Draw a circle touching three converging lines		

c. Biology

Unit	SLO	Package Code	Package Name
GASEOUS EXCHANGE	Differentiate among respiration, gas exchange and breathing.	G10BU10P01	Gaseous Breathing Exchange:
	Describe the process of gaseous exchange in plants by comparing photosynthesis and respiration.		
	Describe the mechanism of breathing in term of movements of ribs and diaphragm.		
HOMEOSTASIS	Define homeostasis and describe its importance.	G10BU11P01	Homeostasis and Excretion in Plants
	Describe the mechanisms / adaptations in plants for the excretion / storage of CO ₂ , H ₂ O, O ₂ , latex, resins and gums.		
	Explain osmotic adjustments in plants		
		G10BU11P02	Role of Human Organs in Homeostasis
	Explain the role of skin in regulating body temperature.		
	Describe how lungs keep the carbon dioxide concentration down to certain level.		
	Explain that kidneys control the blood composition.		
	Identify the different organs of urinary system.		
	Relate the structure of kidney with its function.		

	State that main role of kidney is urine formation.		
	Explain that kidney plays an important role in osmoregulation.		
SUPPORT AND MOVEMENT	Differentiate between moveable joints and immovable joints.	G10BU13P02	Joints and Locomotion
	State the role of ligaments and tendons.		
	Describe the location and movement of hinge joints.		
	Identify ball-n-socket joints in human body.		
	Define antagonism.		
	Describe the action of flexors and extensors as a pair of opposing muscles selecting biceps and triceps as example.		
	Describe the effect of by deficiency of calcium on bones and relate this deficiency with osteoporosis.		
	Discuss the causes, symptoms, and treatment of arthritis.		
	Relate the onset of arthritis with age and weight-bearing joints.		
INHERITANCE	Define genetics.	G10BU15P01	Genetics
	Explain how genes control inheritance of characters.		
	Describe the composition of chromatin material		
	Define gene (a localized region of DNA that codes for a protein).		
	State clearly the difference between a gene and an allele.		
	Explain that gene is a unit of inheritance and that it can be copied and passed on to the next generation.		
	Describe complete dominance using the terms dominant, recessive, phenotype, genotype, homozygous, heterozygous, P1, F1, F2 generations and proving it diagrammatically through a monohybrid genetic cross.		

MAN AND HIS ENVIRONMENT	Describe food chains and food webs.	G10BU16P02	Food Chains and Webs
	Relate biogeochemical cycles with flow of energy and ecological balance.		
	Interpret pyramids of numbers and biomass.		
	Describe carbon and nitrogen cycles.		
	Describe the importance of balance in nature.		
	Explain the human impact on environment.	G10BU16P05	Human Impact on the Environment
	Explain conservation of nature.		
	Explain the human impact on environment.	G10BU16P06	Global and Regional Environmental Problems
	Explain some global and regional environmental problems (population growth, urbanization, global warming, deforestation, acid rain).		
	Explain some global and regional environmental problems (population growth, urbanization, global warming, deforestation, acid rain).		
	Explain some global and regional environmental problems (population growth, urbanization, global warming, deforestation, acid rain).		
	Describe levels of ecological organization.	G10BU16P01	Ecosystem
	Define ecosystem.		
	Define ecosystem.		
	Describe components of the ecosystem.		
	Describe the interrelationships between different components of the ecosystem.		

	Explain that the sun is the principal source of energy for all biological systems.		
	Describe and compare energy relations between different trophic levels.		
	Compare and contrast the flow of materials (cyclic) and the flow of energy (non-cyclic) in the ecosystem.		
COORDINATION AND CONTROL	Define reflex action and reflex arc.	G10BU12P02	Central Nervous System
	Trace the path of a nervous impulse in case of a reflex action.		
	Differentiate between the cross sectional views of brain and spinal cord, with reference to white and grey matter.		
REPRODUCTION	Describe different types of asexual reproduction i.e. binary fission, budding, spore formation and vegetative propagation.	G10BU14P01	Asexual Reproduction
	Outline the binary fission, multiple fission, budding and fragmentation as asexual methods of reproduction in animals.		
	Define cloning.		
	Distinguish between vegetative propagation and artificial propagation.		
INHERITANCE	Describe complete dominance using the terms dominant, recessive, phenotype, genotype, homozygous, heterozygous, P1, F1, F2 generations and proving it diagrammatically through a monohybrid genetic cross.	G10BU15P02	Mendel's Laws of Genetics
	Demonstrate that the 3:1 monohybrid F-2 phenotypic ratio is an evidence of segregation of alleles.		
	State Mendel's law of Segregation.		

	Demonstrate that 9:3:3:1 dihybrid F-2 phenotypic ratio is an evidence of independent assortment.		
	State Mendel's law of Independent Assortment.		
	Selecting the example of ABO blood group system, explain co-dominance.		
	Explain incomplete dominance in Japanese 4 O' Clock plant.	G10BU15P03	Dominance and Variation
	Describe the sources of variation.		
	Relate meiosis with variation.		
HOMEOSTASIS	State that nephron is the excretory unit of kidney.	G10BU11P03-A (partial) G10BU11P03-B (partial) G10BU11P03-C	kidney functions
	Locate the different parts of nephrons and relate them with their function.		
	Describe that urine formation involves three processes i.e. filtration, reabsorption and secretion.		
	Identify the causes of kidney stone.		
	Identify lithotripsy and surgery as the methods to remove kidney stones.		
	Outline the causes of kidney failure.		
	Explain that dialysis is one of the treatments in kidney failure.		
PHARMACOLOGY	Categorize and describe the effects of addictive drugs (sedatives, narcotics and hallucinogens).	G10BU18P02 (partial)	Addictive Drugs
	State the associated problems of drugs addictions i.e. severe social abandonment and crimes.		
	Identify the symptoms of addiction.		
COORDINATION AND CONTROL	Explain what coordination means.	G10BU12P01	Types of Coordination Components of Nervous Coordination

	Identify the two main types of coordination in living organisms, i.e., Nervous and Hormonal (chemical).		
	Differentiate between the modes of coordination i.e., electrical in case of nervous and chemical in case of hormonal.		
	Identify the main organs responsible for coordination and control.		
	State that receptors receive stimuli and transmit information to effectors through CNS		
	Define neuron and describe the structure of a general neuron.		
	Name the three types of neuron involved in reflex action.		
SUPPORT AND MOVEMENT	Describe the role of skeleton in support and movement.	G10BU13P01	Human Skeleton
	Describe the main components of the axial skeleton and the appendicular skeleton.		
	Define skeleton and differentiate between cartilage and bone.		
	Explain that skeleton system is actually a dynamic, living tissue that is capable of growth, adapts to stress and repairs itself after injury.		
	Describe major achievements of genetic engineering in curing animal diseases (foot-and-mouth disease, Coccidiosis, Trypanosomiasis) and in animal propagation (animal cloning).	G10BU17P03 (partial)	Benefits of Genetic Modification
	Describe the application of genetic engineering in the production of human insulin and growth hormones		
REPRODUCTION	Describe different organs of the male and female reproductive systems of rabbit.	G10BU14P04-A (partial) G10BU14P04-B (partial)	Sexual Reproduction in Rabbits and Sexually Transmitted Diseases
	Describe the processes of gametogenesis in rabbit.		
	Explain AIDS as an example of sexually transmitted diseases.		

	Define biotechnology and explain its importance.	G10BU17P01	Biotechnology
	Define fermentation.		
	Explain the method of fermentation by yeast and bacteria.		
	Identify different fermentation products and their importance in daily life i.e. yogurt making, bread making, making of cheese and production of alcohol.		
	Relate biotechnology with genetic engineering and fermentation.		
	Explain the use of fermenter in large-scale production of microorganisms and their products.		
	Describe the procedure of using fermenters.		
	Describe the advantages / profitability of using fermenters in preparing medical products.		
REPRODUCTION	Describe sexual reproduction in plants by explaining the life cycle of a flowering plant.	G10BU14P03	Sexual Reproduction in Plants and Animals
	Describe the adaptations in the structure of wind-pollinated and insect-pollinated flowers.		
	Describe the structure of seed.		
	Distinguish between epigeal and hypogeal germination.		
	Describe the conditions necessary for germination of seeds.		
MAN AND HIS ENVIRONMENT	Define fertilization and differentiate between external and internal fertilization.	G10BU16P04	Pollution and its Types Part 2
	Describe effects of pollution on plants, animals and human beings.		
	Describe possible actions to control pollution		
	Explain different strategies for conservation of nature (reduced		

	resource use, reuse and recycling of materials etc.)		
INHERITANCE	Relate meiosis with variation.	G10BU15P04	Variation and Evolution
	Describe variation and explain difference between continuous and discontinuous variation by giving examples like, height, weight, IQ, gender and blood groups in population.		
	Define organic evolution and explain how variation can lead to evolution.		
	Describe how variation leads to competition in a population and differential survival by best fitting the environment.		
	Assess selection as a possible means of Evolution.		
MAN AND HIS ENVIRONMENT	Develop an understanding of artificial selection as a means of improvement of yield in economically important plants, like wheat, rice etc.	G10BU16P03	Pollution and its Types Part 1
	Explain causes of air, water, and land pollution.		
	Describe effects of pollution on plants, animals and human beings.		
PHARMACOLOGY	Define Pharmacology as the detailed study of drugs.	G10BU18P01	Pharmacology
	Define the term 'drug' (the substance or product that is used to modify physiological systems of the body).		
	Enlist the various sources of drugs i.e. minerals, animals, plants, synthetics, microorganisms.		
	Describe the principle usages of painkillers, antibiotics, vaccines and sedatives.		

GASEOUS EXCHANGE	Describe the roles of the parts of air passageway and of lungs.	G10BU10P02-A (partial) G10BU10P02-B (partial) G10BU10P02-C (partial)	Human Respiratory System
	Describe briefly diseases related to respiratory system like bronchitis, emphysema, pneumonia, asthma, and lung cancer.		
	Describe the biological consequences of smoking in relation to the lungs and circulatory system.		
COORDINATION AND CONTROL	Label the diagram of human brain.	G10BU12P03	Types of Nervous System
	Explain the function of these parts of brain; cerebrum, cerebellum, pituitary gland, thalamus, hypothalamus, medulla oblongata.		
	Identify the main organs responsible for coordination and control.	G10BU12P04	Chemical Coordination and The Human Ear
	Define the terms; hormone and endocrine system.		
	Explain how adrenaline may be involved in exercise and emergency conditions and use gained knowledge to apply to different hormones.		
	Outline the parts of endocrine system; major glands of this system (Pituitary, Thyroid, Pancreas, Adrenal, Gonads) and names of their respective hormone.		
	Describe the term “Negative feedback” with reference to Insulin and glucagon.		
	Describe the structure of human auditory and visual receptors.		
		G10BU12P05	The Human Eye
	Explain the role of ear and eye in maintenance of homeostasis through balance and accommodation.		
	Describe the structure of human auditory and visual receptors.		

	Describe the pupil reflex in dim and bright light.		
	State how short and long sightedness can be treated.		
REPRODUCTION	Distinguish between vegetative propagation and artificial propagation.	G10BU14P02	Reproduction in Plants
	Explain vegetative propagation in plants (through stem, suckers and leaves).		
	Describe the two methods of artificial vegetative propagation (stem cuttings and grafting).		
	Describe sexual reproduction in plants by explaining the life cycle of a flowering plant.		
	Describe single-cell protein and its importance.	G10BU17P02	Genetic Engineering
	Describe the significance of single-cell protein in animal feed.		
	State the significance of single-cell protein in human food.		
	Describe how a gene is transplanted.		
	Describe major achievements of genetic engineering with reference to improvement in agricultural crops (herbicide resistance, virus resistance and insect resistance).		

d. Chemistry

Unit	SLO	Package Code	Package Name
Chemical Equilibrium	Define chemical equilibrium in terms of a reversible reaction. (Understanding)	G10CU09P01	Reversible Reactions and Dynamic Equilibrium
	Write both the forward and the reverse reactions and describe the macroscopic characteristics of each. (Applying)		
	State the necessary conditions for equilibrium and the ways that equilibrium can be recognized. (Understanding)		
	Define Law of mass action (Understanding)	G10CU09P02	Law of Mass Action
	Derive an expression for the equilibrium constant and its units (Applying)		
	Write the equilibrium constant expression of a reaction.		
Acids, Bases and Salts	Complete and balance a neutralization reaction. (Applying)	G10CU10P02	Acid-Base Titrations and Amphoteric Nature of Water
	Write the equation for the self-ionization of water. (Remembering)		
	Given the hydrogen ion or hydroxide ion concentration, classify a solution as neutral, acidic, or basic. (Applying)		
Organic Chemistry	Identify some general characteristics of organic compounds. (Remembering)	G10CU11P01	Sources and Uses of Organic Compounds
	Explain the diversity and magnitude of organic compounds. (Understanding)		
	List some sources of organic compound (Applying)		
	List the uses of organic compounds (remembering)		
	Recognize structural, condensed, and molecular formulas of the straight chain hydrocarbons up to ten carbon atoms. (Understanding)		
Environmental Chemistry I: Atmosphere	Describe major air pollutants. (Understanding)	G10CU14P03	Air Pollutants
	Describe sources and effects of air pollutants. (Understanding)		

	Describe sources and effects of air pollutants. (Understanding)		
	Describe acid rain and its effects (Understanding)		
Chemical Industries	Make a list of raw materials for Solvay process. (Applying) Outline the basic reactions of Solvay process. (Analyzing) Develop a flow sheet diagram of Solvay process. (Creating) Describe the composition of urea. (Understanding) Develop a flow sheet diagram for the manufacture of urea. (Creating) List the uses of urea. (Remembering) Define petroleum (Remembering) Describe the formation of petroleum and natural gas. (Understanding) Describe the composition of petroleum. (Remembering) Describe briefly the fractional distillation of petroleum. (Applying)	G10CU16P02	Some Industrial Processes
Environmental Chemistry I: Atmosphere	Define atmosphere. (Remembering) Explain composition of atmosphere. (Understanding) Differentiate between stratosphere and troposphere. (Analyzing) Summarize the components of stratosphere and troposphere. (Understanding)	G10CU14P01	Compositions and Layers of Atmosphere

Biochemistry	Distinguish between mono-, di- and trisaccharides. {Understanding}	G10CU13P01	Carbohydrates and Proteins
	Explain the sources and uses of carbohydrates, proteins, and lipids. {Understanding}		
Environmental Chemistry I: Atmosphere	Describe global warming. (Understanding)	G10CU14P02	Greenhouse Effect, Global Warming, and Ozone Depletion
	Describe ozone depletion and its effects. (Understanding)		
	Explain ozone formation. (Understanding)		
	Describe major air pollutants. (Understanding)		
Environmental Chemistry II: Water	Describe the occurrence of water and its importance in the environment including industry. (Analyzing)	G10CU15P01	Water (Composition, Properties, Sources and Importance)
	Describe the composition and properties of water. (Understanding)		
	Review our dependence on water and the importance of maintaining its quality. (Analyzing)		
	Differentiate among soft, temporary and permanent hard water. (Analyzing)		
Organic Chemistry	Recognize and identify a molecule's functional groups.(Understanding)	G10CU11P02 (partial)	Classification of Organic Compounds
	Differentiate between different organic compounds on the basis of their functional groups. (Analyzing)		
	Define functional group. (Remembering)		
Hydrocarbons	Explain why a systematic method of naming chemical compounds is necessary. {Analyzing}	G10CU12P01-A (partial)	Hydrocarbons and Functional Groups
	Characterize a hydrocarbon. {Understanding}	G10CU12P01-B (partial)	
Biochemistry	Explain the sources and uses of carbohydrates, proteins, and lipids. {Understanding}	G10CU13P02-A (partial)	Lipids, Nucleic Acids, and Vitamins
	Describe the importance of nucleic acids. {Understanding}	G10CU13P02-B (partial)	
Environmental Chemistry II: Water	Describe methods for eliminating temporary and permanent hardness of water. (Applying)	G10CU15P02-A (partial) G10CU15P02-B	Soft and Hard water, Water Pollution

	Identify water pollutants. (Analyzing)	(partial) G10CU15P02-C (partial)	
	Describe industrial wastes and household wastes as water pollutants. (Understanding)		
	Describe the effects of these pollutants on life. (Understanding)		
	Describe the various types of water borne diseases. (Understanding)		
Acids, Bases and Salts	Define and give examples of Arrhenius acids and bases. (Understanding)	G10CU10P01	Three Concepts of Acids and Bases
	Use the Bronstad-Lowry theory to classify substances as acids or bases, or as proton donors or proton acceptors. (Applying)		
	Classify substances as Lewis acids or bases. (Analyzing)		
	Complete and balance a neutralization reaction. (Applying)		
Chemical Industries	Describe some metallurgical operations. (Applying)	G10CU16P01-A(partial)	Metallurgical Operations
		G10CU16P01-B(partial)	
	Describe some metallurgical operations. (Applying)	G10CU16P03-A(partial)	
Hydrocarbons	Write a chemical equation to show the preparation of alkanes from hydrogenation of alkenes and alkynes and reduction of alkyl halides. { Remembering)	G10CU12P02	Preparation and Properties of Hydrocarbons
	Write a chemical equation to show the preparation of alkenes from dehydration of alcohols and dehydrohalogenation of alkyl halides. { Remembering)		
	Write chemical equations showing halogenation for alkanes, alkenes and alkynes. {Remembering)		
	Write chemical equations showing reaction of KMnO. with, alkenes and alkynes. {Remembering)		

Biochemistry	Distinguish between mono-, di- and trisaccharides. {Understanding}	G10CU13P01	Carbohydrates and Proteins
	Explain the sources and uses of carbohydrates, proteins, and lipids. {Understanding}		

e. Physics

Unit	SLO	Package Code	Package Name
Simple harmonic motion and waves	draw forces acting on a displaced pendulum.	G10PU01P01	Waves and Pendulums
	solve problems by using the formula $T = 2\pi \sqrt{l/g}$ for simple pendulum		
	describe wave motion as illustrated by vibrations in rope, slinky spring and by experiments with water waves.		
	identify transverse and longitudinal waves in mechanical media, slinky and springs.		
	describe that waves are means of energy transfer without transfer of matter.		
Simple harmonic motion and waves	distinguish between mechanical and electromagnetic waves.	G10PU01P02	Types of Waves and Wave Behaviour
	define the terms speed (v), frequency (f), wavelength (λ), time period (T), amplitude, crest, trough, cycle, wave front, compression and rarefaction.		
	derive equation $v=f\lambda$.		
	solve problems by applying the relation $f = 1/T$ and $v = f\lambda$.		
	describe properties of waves such as reflection, refraction and diffraction with the help of ripple tank.		
Sound	explain how sound is produced by vibrating sources and that sound waves require a material medium for their propagation.	G010PU02P01	Sound Waves
	describe the longitudinal nature of sound waves (as a		

	series of compressions and rarefactions).		
	define the terms pitch, loudness and quality of sound.		
	describe the effect of change in amplitude on loudness and the effect of change in frequency on pitch of sound.		
	describe how reflection of sound may produce echo.		
Geometrical optics	describe the terms used in reflection including normal, angle of incidence, angle of reflection and state laws of reflection..	G010PU03P01	Reflection and its Terms
	solve problems of image location by spherical mirrors by using mirror formula.		
	define the terminology for the angle of incidence i and angle of refraction r and describe the passage of light through parallel-sided transparent material.		
Geometrical optics		G010PU03P02	Total Internal Reflection
	solve problems by using the equation $\sin i / \sin r = n$ (refractive index).		
	describe how total internal reflection is used in light propagation through optical fibres.		
Geometrical optics	state the conditions for total internal reflection.	G010PU03P03	Lenses and Glass Prism
	describe the passage of light through a glass prism.		
	describe how light is refracted through lenses.		
Geometrical optics	define the terms resolving power and magnifying power.	G010PU03P05	Microscopes, Telescopes, and the Human Eye
	draw ray diagram of simple microscope and mention its magnifying power.		

	draw ray diagram of compound microscope and mention its magnifying power.		
	draw ray diagram of a telescope and mention its magnifying power		
	draw ray diagrams to show the formation of images in the normal eye, a shortsighted eye and a long-sighted eye.		
	describe the correction of short-sight and long-sight		
<i>Electrostatics</i>	describe simple experiments to show the production and detection of electric charge.	G010PU04P01	Static Electricity and Electroscopes
	describe experiments to show electrostatic charging by induction.		
	describe the construction and working principle of electroscope.		
Electrostatics	state that there are positive and negative charges.	G10PU04P03	Electrostatics
	describe the concept of electrostatic potential.		
	define the unit "volt".		
	describe potential difference as energy transfer per unit charge.		
Current electricity	<i>Define electricity and its types, and how the flow of current works</i>	G010PU05P01	Current Electricity
	define electric current.		
	describe the concept of conventional current.		
	understand the potential difference across a circuit component and name its unit .		
Current electricity	describe the use of electrical measuring devices like galvanometer, ammeter and voltmeter (construction and working principles not required).	G010PU05P02	Ohm's Law, Joule's Law, and Electrical Power

	describe Ohm's law and its limitations.		
	define resistance and its unit(Ω).		
	describe the factors affecting the resistances of a metallic conductor.		
	describe how energy is dissipated in a resistance and explain Joule's law.		
	apply the equation $E = I \cdot V \cdot t = I^2 R t = \frac{V^2}{R} t$ to solve numerical problem.		
	calculate the cost of energy when given the cost per kWh.		
<i>Current electricity</i>	distinguish between D.C and A.C.		
<i>Electrostatics</i>	describe that the capacitor is charge storing device.		
<i>Electrostatics</i>	define capacitance and its unit.		
<i>Current electricity</i>	identify circuit components such as switches, resistors, batteries etc.		
<i>Current electricity</i>	construct simple series (single path) and parallel circuits (multiple paths).		
<i>Current electricity</i>	predict the behaviour of light bulbs in series and parallel circuit such as for celebration lights.		
<i>Current electricity</i>	state the functions of the live, neutral and earth wires in the domestic main supply.		
<i>Current electricity</i>	state reason why domestic supplies are connected in parallel.		
<i>Current electricity</i>	describe hazards of electricity (damage insulation, overheating of cables, damp conditions).		
<i>Current electricity</i>			
<i>Current electricity</i>	explain the use of safety measures in household electricity, (fuse, circuit breaker, earth wire).		
<i>Current electricity</i>			

G010PU05P03

Circuits and Hazards of Electricity

Electromagnetism	explain by describing an experiment that an electric current in a conductor produces a magnetic field around it.	G010PU06P03	Electromagnetism, Induction, and Transformers
	describe a simple form of A.C generator.		
	describe mutual induction and state its units.		
	describe the purpose of transformers in A.C circuits.		
	identify that a transformer works on the principle of mutual induction between two coils.		
Introductory electronics	describe the basic principle of CRO and make a list of its uses.	G010PU07P02	Electronics
	differentiate between analogue and digital electronics.		
	state the basic operations of digital electronics.		
	identify and draw the symbols for the logic gates (NOT, OR, AND, NOR and NAND).		
	state the action of the logic gates in truth table form.		
Information and communication technology	describe the simple uses of logic gates.	G010PU08P02	Information and Communication Technology
	explain briefly the transmission of 1. electric signals through wires 2. radiowaves through air 3. light signals through optical fibres		
	describe function and use of fax machine, cell phone, photo phone and computer.		
	make a list of the use of E-mail and internet.		
	describe the use of information storage devices such as audio cassettes, video cassettes, hard discs,		

	floppy, compact discs and flash drive. identify the functions of word processing, data managing, monitoring and controlling.		
Radioactivity	describe the structure of an atom in terms of a nucleus and electrons. describe the composition of the nucleus in terms of protons and neutrons. explain that number of protons in a nucleus distinguishes one element from the other. represent various nuclides by using the symbol of proton number Z, nucleon number A and the nuclide notation X. explain that some nuclei are unstable, give out radiation to get rid of excess energy and are said to be radioactive. describe that the three types of radiation are α, β & γ. state, for radioactive emissions: o their nature o their relative ionizing effects. o their relative penetrating abilities. explain that an element may change into another element when radioactivity occurs. represent changes in the composition of the nucleus by symbolic equations when alpha or beta particles are emitted.	G010PU09P01	The Nucleus and Radioactivity
Radioactivity	describe what are radio isotopes. What makes them useful for various applications?. describe briefly the processes of fission and fusion. describe hazards of radioactive materials.	G010PU09P02	Nuclear Fission and Fusion Hazards of Radioactive Materials

<i>Sound</i>	describe audible frequency range.	G010PU02P02-A(partial)	Ultrasound and its Applications
<i>Electrostatics</i>	describe one situation in which static electricity is dangerous and the precautions taken to ensure that static electricity is discharged safely.	G10PU04P04-A(partial)	Static Electricity
<i>Information and communication technology</i>	describe the components of information technology.	G010PU08P02-A(partial)	Information and Communication Technology
<i>Introductory electronics</i>	explain the process of thermionic emission emitted from a filament.	G010PU07P01-A(partial)	Thermionic Emission
<i>Introductory electronics</i>	describe the simple construction and use of an electron gun as a source of electron beam.		
<i>Introductory electronics</i>	describe the effect of electric field on an electron beam.		
<i>Introductory electronics</i>	describe the effect of magnetic field on an electron beam.		
<i>Radioactivity</i>	describe that radioactive emissions occur randomly over space and time.	G010PU09P03-A(partial) G010PU09P03-B(partial)	Capacitors, Electrostatics, and Radioactive Half-Life
<i>Radioactivity</i>	explain the meaning of half life of a radioactive material.		
<i>Radioactivity</i>			

5. How to Use This Document in Class

This guide is designed to make lesson planning and classroom delivery easier. Teachers can use it in the following ways:

- **Lesson Planning:** Before each lesson, check the tables to see which video(s) cover the relevant SLO(s) for your class.
- **Classroom Instruction:** Play the appropriate video during class to illustrate concepts, reinforce learning, or introduce new topics.
- **Curriculum Alignment:** Ensure videos are paired with textbook units to maintain alignment with the Pakistan National Curriculum.
- **Revision and Reinforcement:** Use videos to review previous lessons or provide extra practice for students who need it.
- **Multi-Grade or Mixed-Level Classes:** Teachers can identify videos suitable for different grade levels and adjust classroom activities accordingly.
- **Assessment:** Each video package has a corresponding assessment, named with the same package code, which can be found in the accompanying Excel sheets. Teachers can use these assessments to check students' understanding immediately after showing the video or during lesson follow-ups.

The document is structured to make it easy to locate videos quickly, so you spend more time teaching and less time searching.

6. Ethical and Copyright Considerations

All digital content provided through this platform is procured by the Ministry of Federal Education and Professional Training and aligns with the Pakistan National Curriculum. When using these resources, teachers should keep the following points in mind:

- **Authorized Use Only:** Videos are for educational purposes within classrooms or official teaching activities. Do not distribute, copy, or upload content to public platforms without permission.
- **Attribution:** When using content in presentations or shared materials, credit the Ministry as the source.
- **Respect for Learners:** Ensure all content is used in a way that supports inclusive and respectful learning environments.

- **No Modification Without Approval:** Do not alter video content, subtitles, or audio tracks unless officially authorized.
- **Data Privacy:** Avoid sharing student information or recordings when using videos in digital or remote learning settings.

Following these guidelines ensures that teachers use the digital content responsibly while maintaining legal and ethical standards.

7. Conclusion

This guide is designed to make using digital learning videos simple, efficient, and aligned with the Pakistan National Curriculum and Student Learning Outcomes. By following the mapping tables and video codes, teachers can quickly identify the right resources for each lesson, support student learning effectively, and plan classroom activities with confidence.

We encourage teachers to explore the videos, pair them with assessments, and integrate them into lesson plans to create engaging and interactive learning experiences.

For any questions, technical issues, or additional support, please contact the Federal Directorate of Education (FDE)'s content team or the designated focal points.

Empower your students, explore the digital resources, and make every lesson an engaging step toward learning success.

